

Marketing Mix Model Variables Explained

Time & Base Variables

- **ds**: Date/timestamp column for time series analysis
- **dep_var**: Dependent variable (sales/conversions being predicted)
- **weekday**: Day of week indicator

Japanese Seasonal Events

- **GT**: Golden Time (seasonal shopping period)
- **CP230317**: Campaign period starting March 17, 2023
- **CP230421**: Campaign period starting April 21, 2023
- **CP230811**: Campaign period starting August 11, 2023
- **obon230812**: Obon Festival period (August 12, 2023)
- **GW230429**: Golden Week period (April 29, 2023)
- **GW230430**: Golden Week period (April 30, 2023)

Marketing Channels

- **SEM**: Search Engine Marketing campaigns
- **ADNW1**: Advertising Network 1
- **ADNW2**: Advertising Network 2
- **ADNW3**: Advertising Network 3
- **SNS**: Social Network Sites marketing
- **AFF**: Affiliate marketing programs
- **DOUGA**: Video advertising campaigns
- **BRAND**: Brand marketing initiatives

Specific Marketing Activities

- **GAMEPR230324**: Game promotion on March 24, 2023
- **GAMEPR230727**: Game promotion on July 27, 2023
- **INFLU230402**: Influencer campaign on April 2, 2023
- **INFLU230404**: Influencer campaign on April 4, 2023
- **INFLU230408**: Influencer campaign on April 8, 2023
- **INFLU230821**: Influencer campaign on August 21, 2023

Model Output Variables

- **intercept:** Baseline performance without marketing impact
- **depVarHat:** Model's predicted values
- **solid:** Solution identifier for this model run
- **cluster:** Solution cluster grouping
- **top_sol:** Indicator of top-performing solution